

Eric Son

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EDUCATION

Columbia University

Bachelor of Science in Computer Science

New York, NY

Expected Graduation: May 2028

- **GPA:** 3.76/4.00
- **Relevant Coursework:** Data Structures, Advanced Programming, AI, Intro to Databases, C++ Deep Dive

EXPERIENCE

Infosys

Jun 2026 – Aug 2026

Software Engineering Intern

Bangalore, India

- Built a health-monitoring dashboard with SQLite backend, polling 20 internal services across Infosys Living Labs every 30s and replacing manual status checks with real-time uptime/latency tracking for 100+ engineers
- Delivered root-cause summaries within 1 minute of failure detection by engineering an automated incident-response pipeline using NVIDIA Nemotron to analyze GitHub code commit diffs across production outages

Cirque

Aug 2025 – Dec 2025

Software Engineering Intern

New York, NY

- Built a React Native mobile wardrobe app with Supabase JWT-based authentication and user-scoped cloud storage, enabling photo import/storage, virtual outfit visualization, and tagging/filtering across 5+ clothing categories
- Modularized a static React.js frontend with dynamic routing and code splitting, cutting load times by 25%

Ideal Course Mapping VIP (Vertically Integrated Project)

Jan 2025 – May 2025

Undergraduate Research Intern – Georgia Tech

Atlanta, GA

- Designed a course-planning tool using topological sort given a student's major, credits, grade, and prerequisites
- Optimized client-side data fetching with Axios caching and skeleton loading, reducing app latency by 20%

PROJECTS

CourtVision | React, TypeScript, FastAPI, PyTorch, OpenCV, PostgreSQL

Sep 2026 – Present

- Developing a computer vision tennis analytics platform that processes single-camera match footage to track players and ball trajectories, segment rallies, and generate shot placement and player movement heatmaps analytics
- Building an asynchronous video processing pipeline with FastAPI, PyTorch, and OpenCV to detect tennis court geometry and transform tracked pixel coordinates into normalized court positions using homography

Token Optimizer | React, TypeScript, Node.js

Aug 2026 – Sep 2026

- Built a proxy layer for Cursor and its LLM backend to intercept and compress tool-call payloads before reaching the model, cutting overall context usage by 30% and reducing estimated per-session API cost by 10%
- Adding compaction passes to minify JSON, strip code comments, and remove filler language from prompts

HTTP Web Server | C, Git, UNIX

Sep 2025 – Dec 2025

- Engineered an HTTP/1.0-compliant TCP web server using the Berkeley sockets API, handling request parsing, static file serving, directory-index resolution, protocol-correct status codes, and path-traversal protection

2D Cell Segmentation | Python, Pytorch

Sep 2025 – Oct 2025

- Developed a 2D U-Net model for single-cell segmentation using 7,590 multi-channel fluorescence microscopy images of baby hamster kidneys; achieved 98.4% test accuracy using supervised training with binary ground-truth masks

LEADERSHIP

Teaching Assistant (COMS3157)

Sep 2026 – Present

Columbia CS Department

New York, NY

- Reinforced understanding of C, memory management, and Unix/networking fundamentals for 250+ students

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/HTML/CSS, Bash, TypeScript, SQL

Software/Frameworks: React, React Native, Node.js, Express.js, Linux/UNIX, Git, CI/CD

Developer Tools: Postman, Figma, Jira, Vercel, MongoDB, AWS, Render, Docker

Interests: Basketball, Bowling, Tennis, Classical Music, Anime, Running, Poker